

Table 1: A list of features in 5 categories: bibliometric (Bio), author-related (Auth), venue-related (Ven), statistical (Stat), and semantic (Sem). Citations are added for certain features extracted using machine learning models. S2: Semantic Scholar API (<https://www.semanticscholar.org/product/api>). Local: local implementations of algorithms or models. Flag: citation-based features are marked with an open circle $\circ$.

Type	Flag	Feature Name	Description	Type	Source
Bib	$\circ$	#citations	how many time the paper is cited	integer	S2
		normalized citations	average citations per year since the paper was published	integer	S2
		citation velocity	average citations for the last 3 years	integer	S2
		citation next	number of citations in the first 3 years after publication	integer	S2
		#references	number of references the target paper cites	integer	S2
		self-citations ratio	number of self-citations divided by the total number of references	float	S2
		openaccessflag	whether the paper has open access	binary	S2
		age	number of years since the paper was published	integer	S2
		coCite2	number of co-cited papers within 2 years after the target paper was published	integer	S2+Local
		coCite3	number of co-cited papers within 3 years after the target paper was published	integer	S2+Local
		#influential citation	number of influential citations the paper receives	integer	S2
		#influential reference	number of influential references the paper cites	integer	S2
Auth		u rank	normalized university rank of the first author	float	Times
		author count	number of authors of the target paper	integer	S2+Local
		avg pub	average number of publications of authors	float	S2+Local
		avg hidx	average h-index of authors	float	S2+Local
		avg auth cites	average number of citations of authors	float	S2+Local
	$\circ$	avg high inf cites	average number of highly influential citations of all authors	float	S2+Local
Venue		Venue CiteScore	a measure of citation impact for journals	float	Scopus
		Venue SNIP	Source Normalized Impact per Paper aggregated for Journal	float	Scopus
		Venue Scholarly Output	number of papers published in a certain venue in the past 3 years	float	Scopus
		Venue Percent Cited	proportion of documents that have received at least 1 citation	float	Scopus
		Venue Citation Count	citations received in one year for the paper published in the previous 3 years	integer	Scopus
		SJR	SCImago Journal Rank	float	Scopus
		Venue subject code	subject category of the journal	integer	Elsevier
Stat		real p	minimum p -value of all the p -values extracted	float	Local
		real p sign	signs parsed from p -value expressions	binary	Local
		p val range	the difference of the highest and the lowest p -value in the paper	float	Local
		num hypo tested	number of hypothesis tests	integer	Local
		extend p	whether the p -value features are associated with a test	binary	Local
		num significant	the total number of significant p -values	integer	Local
		sample size	the number of participants or observations	integer	Local
Sem		upstream influential methodology count	influential references classified as methodology based on citation context	integer	S2
		funded	whether an acknowledgment of a funding agency is recognized [3]	binary	Local
		reference background	number of reference papers cited as background	integer	S2
		reference methodology	number of reference papers cited as methodology	integer	S2
		reference result	number of reference papers cited as result	integer	S2
	$\circ$	citations background	times this paper cited as background	integer	S2
	$\circ$	citations methodology	times this paper cited as methodology	integer	S2
	$\circ$	citations result	times this paper cited as result	integer	S2
		score1	concept-pair-based innovation score [17]	integer	Local
		gender index	gender diversity index of all authors	float	API
		table	number of tables	integer	Local
		figure	number of figures	integer	Local
		reading score	readability of the claim calculated by Flesch-Kincaid method	float	Local
		subjectivity	how subjective is the claim	float	AllenNLP
		sentiment	sentiment analysis for claims	binary	AllenNLP
		theory count	number of theory/model names in the main text [5]	integer	Local

Venue-level features. Venue-level features reflect the overall quality of the journal where the target paper is published. Intuitively, high-ranking journals, which tend to have higher impact factors on average, potentially publish higher-quality papers, which are potentially correlated with replicability. Venue-level features are retrieved from the Scopus API¹, and a journal handbook with information on more than 40,000 journals indexed by Elsevier. Although certain venue-level features are citation-based and may also change with time, the change is relatively slow [1] compared with paper-level features, so they are not excluded.

Statistical features. In SBS papers, many hypotheses are tested by statistical methods, and the significance of the testing results is indicated by p -values. We extract the p -values, which are potential indicators of the likelihood that claims can be successfully replicated, using a heuristic method [4] based on regular expressions.

Semantic features. This category contains 16 features extracted from the abstract and body text of the paper. These features were explored because details such as citation context functions, acknowledgments, and readability may contain cues about replicability. Here, we elaborate on our rationale for including several features and describe how they are extracted.

¹<https://www.scopus.com/sources>